National University of Computer and Emerging Sciences

Agentic AI CS-4015 (Section: A, B, C)

Course Project PHASE II

Due Date: 19th April 2026, Sunday

THE STUDIO FLOOR:

Video and Audio Synthesis Layer

■ Introduction

Phase 2 transforms the structured outputs of Phase 1 into **multimodal audiovisual content**, enabling the transition from abstract narrative to **perceptual media**.

This phase implements a **parallel multi-agent execution architecture**, where independent agents:

- Process audio and video streams concurrently
- Synchronize outputs through temporal alignment
- Maintain state continuity via shared memory

This phase directly satisfies the **core requirement** of generating synchronized facial geometry and audio waveform using multimodal inputs.

■ System Objectives

- Convert *scene_manifest.json* -> audiovisual scenes
- Generate:
  - Voice (TTS / cloning)
  - Video frames
  - Face-mapped characters
  - Lip-synced outputs
- Outputs:
  - *raw_scenes/scene_01.mp4*
  - Audio tracks (.wav)
  - Intermediate frame sequences

■ Architectural Design Principles

3.1 Parallel Processing Architecture

This phase introduces **parallel execution:**

- Audio branch
- Video branch

Both converge at the **Lip Sync Agent**

3.2 Task Graph-based Execution

All scene processing is decomposed using:

- *get_task_graph* (MCP Tool)

Each scene becomes:

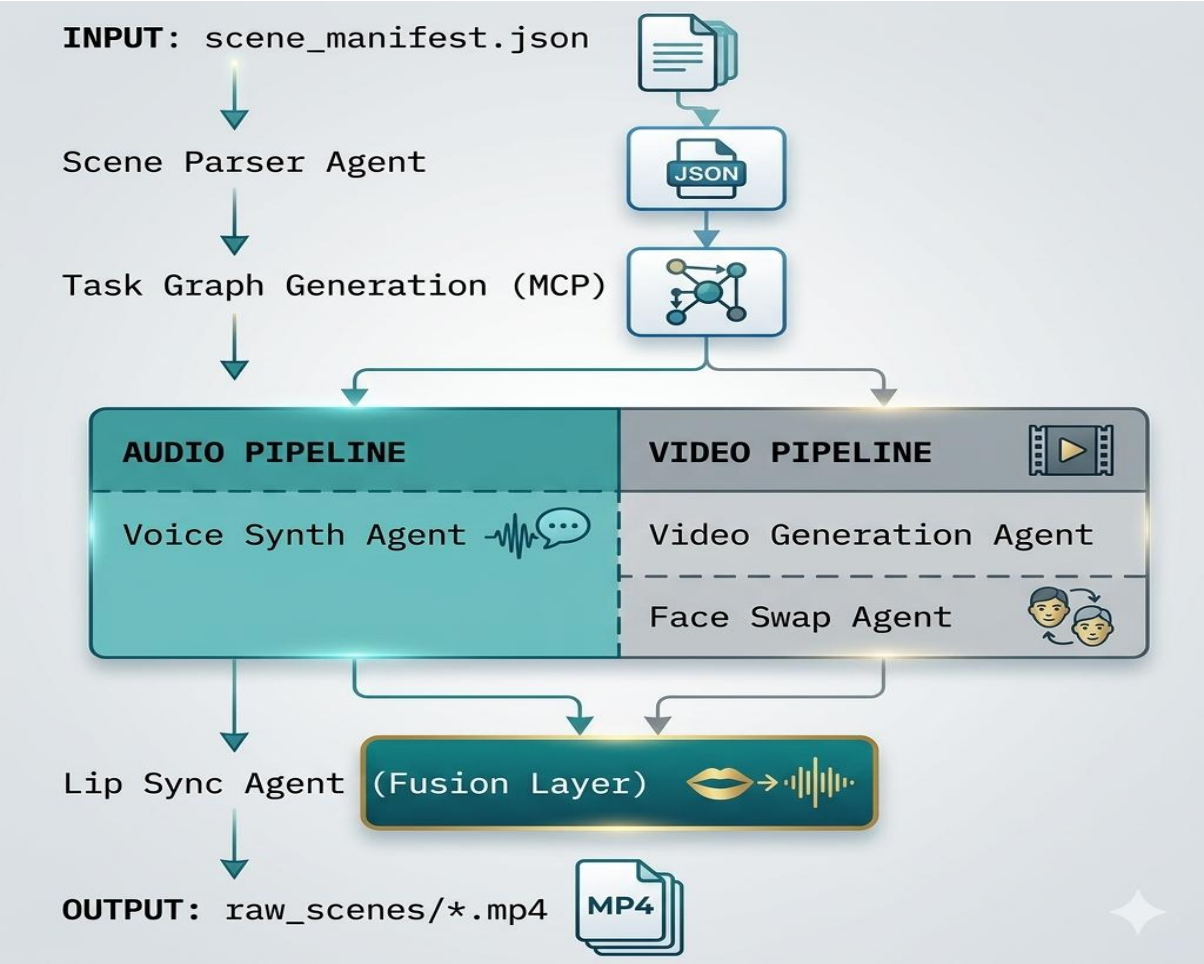
- Independent unit
- Parallelizable workload

3.3 Stateful Resumability

Using *commit_memory*, the system:

- Saves intermediate outputs
- Supports recovery in case of:
  - Failure
  - Interruptions

■ System Architecture (High-Level Flow)



■ Agents Definitions

5.1 Scene Parser Agent

Role:

Transform *scene_manifest.json* into executable tasks.

Core Responsibilities:

- Scene segmentation
- Task distribution
- Parallel routing

MCP Tools:

- *get_task_graph*
- *commit_memory*

5.2 Voice Synthesis Agent

Role:

Generates speech aligned with character identity.

Key Features:

- Voice cloning from reference audio
- Emotion-aware speech synthesis

Reasoning Flow:

- Retrieve character voice profile
- Map dialogue -> speech
- Generate waveform

MCP Tool:

- *voice_cloning_synthesizer*

5.3 Video Generation Agent

Role:

Generates scene visuals from:

- Character references
- Scene descriptions

Inputs:

- Visual cues
- Environment descriptions

MCP Tool:

- *query_stock_footage*

5.4 Face Swap Agent

Role:

Maps generated characters onto video frames.

Critical Constraints:

- Must validate identity before mapping

MCP Tools:

- *face_swapper*
- *identity_validator*

5.5 Lip Sync Agent

Role:

Synchronizes:

- Audio waveform
- Facial movements

Core functionality:

- Frame-by-frame alignment
- Temporal consistency

MCP Tool:

- *lip_sync_aligner*

■ MultiModal Synchronization

This phase solves a key challenge of **Temporal Alignment Problem.**

Ensures:

- Speech timing = lip motion
- Scene continuity

Handled via:

- Lip Sync Agent
- Frame alignment models

■ LangGraph Workflow Design

LangGraph uses:

- Send() API for branching

Nodes:

- *Scene_parser_node*
- *Voice_synth_node*
- *Video_gen_node*
- *Face_swap_node*
- *Lip_sync_node*

■ Deliverables

- *raw_scenes/*.mp4*
- Audio files
- Task graph logs

■ Evaluation Rubric (Phase 2)

Criteria	**Description**	**Marks**
Parallel Architecture	Proper branching implementation	10
Audio Quality	Natural speech synthesis	20
Video Quality	Visual coherence	20
Lip Sync Accuracy	Temporal alignment	10
MCP Tool Usage	Correct integration	5
Fault Tolerance	Resumability	5
Total		**70**